how can I measure deviations from gaussianity of a

ChatGPT Get Plus ×

Measuring **deviations from Gaussianity** in a stellar image is a classic task in PSF (point spread function) analysis, useful for diagnosing optics, guiding errors, atmospheric turbulence, detector issues, or deblending problems.

Below is a practical hierarchy of methods, from simplest diagnostics to more rigorous statistics.

---

1. Fit a 2D Gaussian and examine residuals

Model the stellar image intensity as:

$$I(x,y) = A \exp\left(-\frac{1}{2}\left[\frac{(x-x_0)^2}{\sigma_x^2} + \frac{(y-y_0)^2}{\sigma_y^2} + 2\rho\frac{(x-x_0)(y-y_0)}{\sigma_x\sigma_y}\right]\right) + B$$

Then compute residuals:

$$R(x,y) = I_{\text{data}} - I_{\text{Gaussian fit}}$$

Diagnostics:

- structured residuals → coma, astigmatism, tracking error
- ring residuals → Airy diffraction or defocus
- asymmetric residuals → optical misalignment
- elongated residuals → wind shake or guiding drift

Common scalar summary:

$$\chi_\nu^2 = \frac{1}{N-p}\sum \frac{R(x,y)^2}{\sigma_n^2}$$

Large reduced χ^2 indicates non-Gaussian structure.

---

2. Higher-order moments (kurtosis & skewness)

Compute central image moments:

$$M_{pq} = \sum I(x,y)(x-x_0)^p(y-y_0)^q$$

Useful indicators:

Skewness (asymmetry)

Detects coma or trailing:

$$\gamma_x = \frac{M_{30}}{M_{20}^{3/2}}, \quad \gamma_y = \frac{M_{03}}{M_{02}^{3/2}}$$

Kurtosis (wing strength)

Detects heavy tails or diffraction structure:

$$\kappa_x = \frac{M_{40}}{M_{20}^2} - 3$$

Gaussian expectation:

- skewness = 0
- kurtosis = 0

Heavy PSF wings → positive kurtosis.

---

3. Radial profile comparison

Compute radial intensity profile:

$$I(r)$$

Compare to Gaussian expectation:

$$I_G(r) = I_0 e^{-r^2/2\sigma^2}$$

Useful scalar diagnostics:

Wing excess

$$W = \frac{\int_{r>2\sigma} I(r)\,dr}{\int I(r)\,dr}$$

Gaussianity ratio

Compare FWHM to second moment:

For a Gaussian:

$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma$$

Deviation indicates non-Gaussian wings or core sharpening.

4. Shapelet or Gauss–Hermite expansion

Expand PSF as perturbations around a Gaussian:

$$I(x) = G(x)\left[1 + h_3 H_3(x) + h_4 H_4(x) + \dots\right]$$

Where:

- H_3 captures asymmetry
- H_4 captures peakiness vs heavy tails

Common scalar metric:

$$D = \sqrt{h_3^2 + h_4^2}$$

Widely used in galaxy morphology and PSF modeling.

---

5. Compare with alternative analytic PSF models

Fit multiple models and compare likelihood:

Moffat profile

Better captures atmospheric PSFs:

$$I(r) = I_0 \left(1 + \frac{r^2}{\alpha^2}\right)^{-\beta}$$

β controls wing strength.

Indicators of non-Gaussianity:

- Gaussian β → ∞
- typical seeing β ≈ 2–5

Large improvement in χ^2 over Gaussian → strong deviation.

---

6. Encircled energy curve

Compute fraction of flux inside radius:

$$E(r) = \frac{\int_0^r I(r') 2\pi r' dr'}{\int_0^\infty I(r') 2\pi r' dr'}$$

Compare to Gaussian expectation:

- slower rise → heavy wings
- faster rise → sharper core

---

7. Practical "single-number" metrics

Common summary statistics:

metric	interpretation
reduced χ^2	overall deviation
kurtosis	heavy wings
skewness	asymmetry
Moffat β	wing strength
FWHM / σ ratio	profile mismatch
h_3, h_4	structured deviations
flux outside 2σ	wing excess

---

8. Typical workflow in astronomy pipelines

Typical PSF-quality diagnostics:

1. centroid star
2. normalize flux
3. fit elliptical Gaussian
4. compute residual RMS
5. compute kurtosis
6. optionally fit Moffat
7. flag outliers

Packages commonly used:

- photutils
- astropy.modeling
- galsim
- PSFEx
- DAOPhot

If useful, I can also provide:

- python code snippet for computing Gaussianity metrics
- recommended thresholds for survey pipelines
- comparison of Gaussian vs Moffat fits
- visualization diagnostics used in LSST / HSC pipelines
- robust statistics for undersampled PSFs

B&H Photo · Sponsored

ZWO ASIAIR Plus Control Hub
Run your whole astro rig from your phone—plan, align, shoot.

Ads do not influence the answers you get from ChatGPT. Your chats stay private.
Learn about ads and personalization >